

Objective Mackey and Tambara functors

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Abstract (and Object)

- ▶ The first word in the talk title is intended in the sense that finite sets are objective natural numbers. Natural numbers are an abstraction from sets.
- ▶ At the objective level, the axioms defining abstract Mackey and Tambara functors are categorically familiar.
- ▶ The first step was taken by Harald Lindner in 1976 when he recognized that Mackey functors, defined as pairs of functors, were single functors with domain a category of spans.
- ▶ We look at the objective Mackey functor of Mackey functors and how Tambara functors are, or maybe are not, related.

Spans

- ▶ We start with a lexextensive category $\mathcal{E}$.
- ▶ Form the bicategory $\mathrm{Spn}\mathcal{E}$ with the same objects as $\mathcal{E}$ and hom categories $\mathrm{Spn}\mathcal{E}(X, Y) \cong \mathcal{E}_{/X \times Y}$ whose objects are written as spans $X \xleftarrow{u} S \xrightarrow{v} Y$ from X to Y in $\mathcal{E}$. Composition is by pullback.
- ▶ We write $\mathrm{clSpn}\mathcal{E}$ for the classifying category of $\mathrm{Spn}\mathcal{E}$; the morphisms are isomorphism classes of spans.
- ▶ The bicategory $\mathrm{Spn}\mathcal{E}$ is symmetric autonomous monoidal under the tensor product induced by product in $\mathcal{E}$.
- ▶ Each homcategory $\mathrm{Spn}\mathcal{E}(X, Y)$ has finite coproducts preserved by the composition of $\mathrm{Spn}\mathcal{E}$. So $\mathrm{Spn}\mathcal{E}$ is enriched, as a symm. mon. bicategory, in the closed bicategory Cat_+ of categories with finite coproducts and fin.-coprod.-pres. functors; the internal hom $[\mathcal{A}, \mathcal{B}]_+$ of Cat_+ is $\mathrm{Cat}_+(\mathcal{A}, \mathcal{B})$ with pointwise coproduct.
- ▶ The symm. mon. category $\mathrm{clSpn}\mathcal{E}$ is enriched in the usual closed category CMon of commutative monoids.

Mackey functors

- ▶ A *Mackey functor* on $\mathcal{E}$ is a CMon-enriched functor

$$M : \text{clSpn}\mathcal{E} \rightarrow \text{CMon} .$$

- ▶ As an example, let G be a finite group, let $\mathcal{E} = [G, \text{set}]$ be the category of finite G -sets, and let $R : G \rightarrow \text{vect}$ be a finite-dimensional representation of G . Define $M : \text{clSpn}\mathcal{E} \rightarrow \text{CMon}$ by $MX = \text{Set}^G(X, R)$ and, for $\tau : X \rightarrow R$ and $y \in Y$,

$$M(X \xrightarrow{[u, S, v]} Y)(X \xrightarrow{\tau} R)(y) = \sum_{v(s)=y} \tau(u(s)) .$$

- ▶ Another example is the *Burnside functor* $J : \text{clSpn}\mathcal{E} \rightarrow \text{CMon}$ where

$$J(X) = (\text{clSpn}\mathcal{E})(1, X) = (\mathcal{E}/_X)/\cong$$

is the set of isomorphism classes of the slice category $\mathcal{E}/_X$ made into a commutative monoid with addition induced by coproduct.

Green functors

- ▶ Put $\mathbf{Mky}\mathcal{E} = [\mathbf{clSpn}\mathcal{E}, \mathbf{CMon}]_+$, the $\mathbf{CMon}$ -enriched functor category. It is a monoidal $\mathbf{CMon}$ -enriched category by Day convolution.
- ▶ The monoids for this convolution structure are *Green functors on $\mathcal{E}$* . We write $\mathbf{Grn}\mathcal{E}$ for the $\mathbf{CMon}$ -enriched category of Green functors. The Burnside functor J is an example.
- ▶ Let $\mathbf{Lxt}$ denote the 2-category of lexensive categories and functors which preserve finite coproducts and pullbacks. Then we have a pseudofunctor $\mathbf{Mky} : \mathbf{Lxt} \rightarrow \mathbf{Cat}_+$ taking $\mathcal{E}$ to $\mathbf{Mky}\mathcal{E}$ regarded as a category with finite coproducts.

Objective Mackey

- ▶ An *objective Mackey functor* on $\mathcal{E}$ is a Cat_+ -enriched pseudofunctor

$$\mathbb{M} : \text{Spn}\mathcal{E} \rightarrow \text{Cat}_+ .$$

- ▶ Put

$$\mathcal{Mky}\mathcal{E} = \text{Ps}[\text{Spn}\mathcal{E}, \text{Cat}_+]_+ ,$$

the Cat_+ -enriched pseudofunctor bicategory. There is a convolution monoidal structure $\mathbb{M} \star \mathbb{N}$ on $\mathcal{Mky}\mathcal{E}$ characterized by the existence of a pseudonatural equivalence

$$\mathcal{Mky}\mathcal{E}(\mathbb{M} \star \mathbb{N}, \mathbb{L}) \simeq \int_{X, Y} [\mathbb{M}X \otimes \mathbb{N}Y, \mathbb{L}(X \times Y)]_+$$

where the integral denotes the pseudo-end.

Objective Green

- ▶ The monoidales (pseudomonoids) for the convolution monoidal structure $\mathbb{M} \star \mathbb{N}$ we call *objective Green functors on $\mathcal{E}$* and write $\mathbb{Grn}\mathcal{E}$ for the $\mathbf{Cat}_+$ -enriched bicategory they, with the strong monoidal morphisms, form.
- ▶ An objective Green functor is *symmetric* when it is symmetric as a monoidale.
- ▶ An example of an objective Green functor is the objective Burnside functor $\mathbb{J} : \mathbf{Spn}\mathcal{E} \rightarrow \mathbf{Cat}_+$ whose values are slice categories

$$\mathbb{J}(X) = (\mathbf{Spn}\mathcal{E})(1, X) = \mathcal{E}_{/X} ;$$

but it is also useful to notice that $\mathbb{J}$ lands in the sub-2-category $\mathbf{Lxt}$ of $\mathbf{Cat}_+$.

The objective Green of Mackeys

Proposition

The composite pseudofunctor

$$\mathbb{M}ky_{\mathcal{E}} := \mathbf{Spn}\mathcal{E} \xrightarrow{\mathbb{J}} \mathbf{Lxt} \xrightarrow{\mathbb{M}ky} \mathbf{Cat}_+$$

becomes a symmetric objective Green functor when equipped with the multiplication induced by the finite-coproduct-preserving-in-each-variable functors

$$\mathbb{M}ky(\mathcal{E}_X) \times \mathbb{M}ky(\mathcal{E}_Y) \rightarrow \mathbb{M}ky(\mathcal{E}_{X \times Y})$$

taking (M, N) to $M \boxtimes N$ where

$$(M \boxtimes N)(S \xrightarrow{(a,b)} X \times Y) = M(S \xrightarrow{a} X) \otimes N(S \xrightarrow{b} Y) .$$

What's going on here?

- ▶ Math Reviews asked me to review the paper:
David Chan (2024) *Bi-incomplete Tambara functors as $\mathcal{O}$ -commutative monoids*
- ▶ This led me to do some background reading including:
Hill & Hopkins (2016) *Equivariant symmetric monoidal structures*
Blumberg & Hill (2018) *Incomplete Tambara functors*
- ▶ I wrote the review in response to suggestive terminology in those papers. My guess was, if their terminology were to have any justification, that commutative monoids in the symmetric monoidal $\mathbb{M}k_{\mathcal{E}}$ in the symmetric monoidal bicategory $\mathcal{M}k_{\mathcal{E}}$ are related to Tambara functors on $\mathcal{E}$.
- ▶ I cannot yet see this. However, my investigations continue.
- ▶ So let us plug on!

Polynomials

- Given our lex extensive category $\mathcal{E}$, there is a bicategory $\text{Ply}\mathcal{E}$ of polynomials in $\mathcal{E}$; in terminology of [134] Proposition 8.4 it is the bicategory $\text{PolySpn}\mathcal{E}$. The objects are those of $\mathcal{E}$. The morphisms from X to Y are *polynomials* $p = (X \xleftarrow{r} A \xrightarrow{n} B \xrightarrow{t} Y)$ in $\mathcal{E}$. The morphisms between polynomials are commutative diagrams

$$\begin{array}{ccccc}
 & & A & \xrightarrow{n} & B \\
 & \swarrow r & \uparrow \ell & & \searrow t \\
 X & & P & \xrightarrow{\bar{n}} & B \\
 & \nwarrow r' & \downarrow \bar{g} & \lrcorner & \downarrow g \\
 & & A' & \xrightarrow{n'} & B'
 \end{array}$$

$\begin{array}{ccc} & & Y \\ & \nearrow t' & \\ & & \end{array}$

wherein the square as indicated is a pullback. Composition uses bipullback of spans of spans.

- ▶ Finite coproducts exist in each $\text{Ply}^{\mathcal{E}}(X, Y)$ and are preserved by composition with morphisms making the bicategory $\text{Ply}^{\mathcal{E}}$ enriched in Cat_+ .
- ▶ The philosophy of [134] is that it is useful to regard p as a span

$$p = (X \xleftarrow{(n,A,r)} B \xrightarrow{(1_B,B,t)} Y)$$

in $\text{Spn}^{\mathcal{E}}$. Then the morphisms interpret as diagrams

$$\begin{array}{ccccc}
 & & B & & \\
 & \swarrow (n,A,r) & \downarrow (1_B,B,g) & \searrow (1,B,t) & \\
 X & \xleftarrow{(n',A',r')} & B' & \xrightarrow{(1,B',t')} & Y
 \end{array}$$

$\lambda \cdot (1, B, g) =$

Calibrated bicategories

Definition

A class $\mathcal{P}$ of morphisms, whose members are called *neat* (“propres” in French), in a bicategory $\mathcal{M}$ is called a *calibration of $\mathcal{M}$* when it satisfies the following conditions

- P0. all equivalences are neat and, if p is neat and there exists an invertible 2-cell $p \cong q$, then q is neat;
- P1. for all neat p , the composite $p \circ q$ is neat if and only if q is neat;
- P2. every neat morphism is a groupoid fibration;
- P3. every cospan of the form $S \xrightarrow{p} Y \xleftarrow{n} T$, with n a right lifter and p neat, has a bipullback $S \xleftarrow{\tilde{n}} P \xrightarrow{\tilde{p}} T$ in $\mathcal{M}$ with $\tilde{p}$ neat.

A bicategory equipped with a calibration is called *calibrated*. A bicategory $\mathcal{M}$ is called *polynomic* when the class GF of all groupoid fibrations is a calibration of $\mathcal{M}$.

If $\mathcal{C}$ is a finitely complete category then the bicategory $\mathrm{Spn}\mathcal{C}$ is polynomic.

Polynomials in a calibrated bicategory $\mathcal{M} = (\mathcal{M}, \mathcal{P})$

Definition

A *polynomial* (m, S, p) from X to Y in $\mathcal{M}$ is a span $X \xleftarrow{m} S \xrightarrow{p} Y$ in $\mathcal{M}$ with m a right lifter and p neat. A *lax polynomial morphism* $(\lambda, h, \rho) : (m, S, p) \rightarrow (m', S', p')$ is a diagram

$$\begin{array}{ccccc}
 & & S & & \\
 & \nearrow m & \downarrow h & \nwarrow p & \\
 X & \xleftarrow{\lambda} & S' & \xrightarrow{\rho} & Y \\
 & \xleftarrow{m'} & & \xrightarrow{p'} &
 \end{array} \tag{1}$$

in which λ and ρ are 2-cells. We call (λ, h, ρ) *left pseudo*, *right pseudo*, *pseudo* when (respectively) λ , ρ , both are invertible. A 2-cell $\sigma : h \Rightarrow k : (m, S, p) \rightarrow (m', S', p')$ is a 2-cell $\sigma : h \Rightarrow k : S \rightarrow S'$ in $\mathcal{M}$ compatible with λ and ρ . Write $\text{Ply}_{\text{lax}} \mathcal{M}(X, Y)$ for the bicategory of polynomials from X to Y with the lax morphisms.

Adjoint lax morphisms

Proposition

The lax polynomial morphism $(\lambda, h, \rho) : (m, S, p) \rightarrow (m', S', p')$ has a right adjoint in the bicategory $\text{Ply}_{\text{lax}} \mathcal{M}(X, Y)$ if and only if, in $\mathcal{M}$, h has a right adjoint h^ , and λ and the mate $\hat{\rho} : ph^* \Rightarrow p'$ of ρ under $h \dashv h^*$ are invertible.*

Notice that, if the morphism is right pseudo, its right adjoint also will be, and the morphism itself will be pseudo.

Prove it!

Proof.

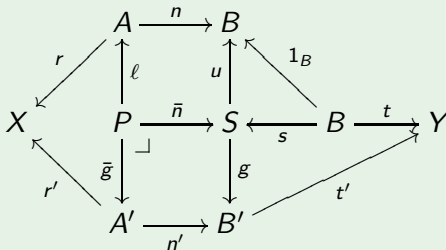
- ▶ There is a forgetful pseudofunctor $\text{Ply}_{\text{lax}} \mathcal{M}(X, Y) \longrightarrow \mathcal{M}$ taking $(\lambda, h, \rho) : (m, S, p) \rightarrow (m', S', p')$ to $h : S \rightarrow S'$. So if $(\lambda, h, \rho) \dashv (\lambda', h^*, \rho')$ then $h \dashv h^*$ and the counit and unit 2-cells $\varepsilon : hh^* \rightarrow 1$ and $\eta : 1 \rightarrow h^*h$ become 2-cells in $\text{Ply}_{\text{lax}} \mathcal{M}(X, Y)$.
- ▶ The mates of $\lambda' : mh^* \rightarrow m'$ and $\rho' : p' \rightarrow ph^*$ are $\lambda'' = (m \xrightarrow{m\eta} mh^*h \xrightarrow{\lambda'h} m'h)$ and $\rho'' = (ph^* \xrightarrow{\rho h^*} p'hh^* \xrightarrow{p'\varepsilon} p')$.
- ▶ The conditions that η should be a polynomial 2-cell are $\lambda \circ \lambda'' = 1_m$ and $\rho'h \circ \rho = p\eta$. So, using the snake identities, $1_{ph^*} = ph^*\varepsilon \circ p\eta h^* = ph^*\varepsilon \circ \rho'h h^* \circ \rho h^* = \rho' \circ p\varepsilon \circ \rho h^* = \rho' \circ \rho''$.
- ▶ The conditions that ε should be a polynomial 2-cell are $\lambda' \circ \lambda h^* = m'\varepsilon$ and $\rho''\rho' = 1_{p'}$. The first of these, again using a snake identity, gives $\lambda''\lambda = 1_{m'h}$.

The result follows. □

The case of $\mathcal{M} = \text{Spn}\mathcal{E}$

Example

The lax morphisms between polynomials here amount to commutative diagrams



wherein the square as indicated is a pullback. It is right pseudo when u is invertible. It is left pseudo when ℓ is invertible.

The tricategory of polynomials

Proposition

There is a tricategory $\text{Ply}_{\text{lax}}\mathcal{M}$ of polynomials in a calibrated bicategory $\mathcal{M}$. The objects are those of $\mathcal{M}$, the hom bicategories are the $\text{Ply}_{\text{lax}}\mathcal{M}(X, Y)$. Composition is given by the pseudofunctors

$$\circ : \text{Ply}_{\text{lax}}\mathcal{M}(Y, Z) \times \text{Ply}_{\text{lax}}\mathcal{M}(X, Y) \longrightarrow \text{Ply}_{\text{lax}}\mathcal{M}(X, Z)$$

which are defined as in the right pseudo case given in [134]. The vertical and horizontal stacking properties of bipullbacks provide the associativity equivalences.

Coproductivity

Suppose $\mathcal{M}$ is a bicategory with finite bicategorical coproducts and such that each hom category $\mathcal{M}(U, V)$ has finite coproducts preserved by composition on each side. It follows [85] that the coproducts are also products where the product projections are right adjoint to the coproduct injections.

Definition

Let $\mathcal{M}$ be a bicategory with finite coproducts. A calibration $\mathcal{P}$ is called *coproductive* when a morphism from a finite coproduct is neat if its composites with the coproduct projections are (this includes that each $0 \rightarrow Z$ is neat) and neat morphisms are right lifters.

Coproductivity

Proposition

Suppose $\mathcal{M}$ is a bicategory with finite bicategorical coproducts and such that each hom category $\mathcal{M}(U, V)$ has finite coproducts preserved by composition on each side. Suppose $\mathcal{M}$ is coproductively calibrated. Then the bicategory $\text{Ply}_{\text{lax}}\mathcal{M}(X, Y)$ also has finite bicategorical coproducts and each of its hom categories has finite coproducts preserved by composition on each sides.

Prove it!

Proof.

The construction is obtained from the diagram

$$\begin{array}{ccccc} & & S + S' & & \\ & \swarrow [m, m'] & \downarrow [h, h'] & \searrow [p, p'] & \\ X & \xleftarrow[n]{} & T & \xrightarrow[q]{} & Y \end{array} \quad (2)$$

The diagram shows a triangular structure. At the top is the expression $S + S'$. Below it is a vertical arrow pointing down to T , labeled $[h, h']$. To the left of T is X , and to the right is Y . A horizontal arrow points from X to T , labeled n . Another horizontal arrow points from T to Y , labeled q . There are also diagonal arrows: one from $S + S'$ to X labeled $[m, m']$, and another from $S + S'$ to Y labeled $[p, p']$. Additionally, there are small arrows pointing from $S + S'$ towards the horizontal arrows: one pointing left towards X labeled $[\lambda, \lambda']$, and one pointing right towards Y labeled $[\rho, \rho']$.

where $[m, m']$ is a right lifter since m and m' are and, using coproductivity, $[p, p']$ is neat since p and p' are. $\square$

More on Ply_{lax}

Let Bicat_+ denote the closed tricategory with objects bicategories admitting finite coproducts and morphisms pseudofunctors preserving them.

Corollary

Ply_{lax} is a Bicat_+ -enriched tricategory.

For each neat $p : U \rightarrow V$ in $\mathcal{M}$, we have the polynomial

$$p_* = (U \xleftarrow{1_U} U \xrightarrow{p} V)$$

and, if p is also a right lifter, we have the polynomial

$$p^* = (V \xleftarrow{p} U \xrightarrow{1_U} U) .$$

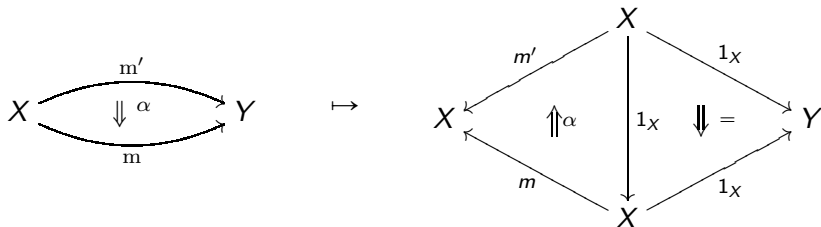
As usual for spans, we have the (bicategorical) adjunction $p_* \dashv p^*$ in Ply_{lax} ; indeed the counit and unit are both pseudo morphisms.

Another adjoint polynomial

Also we have the pseudofunctor

$$(-)^* : \mathcal{M}_{\text{rtlift}}^{\text{coop}} \rightarrow \text{Ply}_{\text{lax}} \mathcal{M}$$

defined by



which is restricted to right lifters m and m' in $\mathcal{M}$.

In particular, if $f \dashv u$ in $\mathcal{M}_{\text{rtlift}}$ then we have the (bicategorical) adjunction $f^* \dashv u^*$ in $\text{Ply}_{\text{lax}} \mathcal{M}$ since pseudofunctors preserve adjunctions.

Let us return now to the case $\mathcal{M} = \mathbf{Spn}\mathcal{E}$.

Every span $U \xleftarrow{m_1} S \xrightarrow{m_2} V$ is a right lifter (since all morphisms in $\mathcal{E}$ including m_1 are powerful).

A span $X \xleftarrow{p_1} B \xrightarrow{p_2} Y$ is neat when p_1 is invertible.

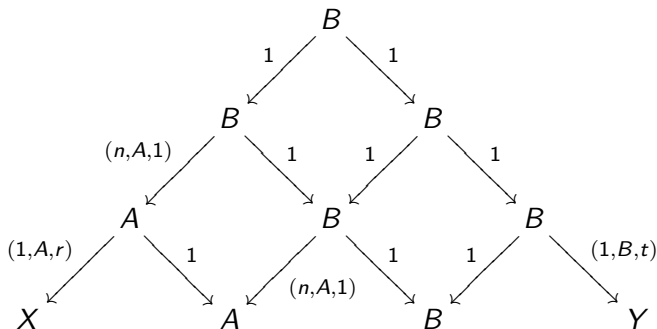
Consequently, for each $f : U \rightarrow V$ in $\mathcal{E}$, we have the three polynomials

$$\begin{aligned} f_{**} &= (U \xleftarrow{1_U} U \xrightarrow{1_U} U \xrightarrow{f} V) = (U \xleftarrow{(1_U, U, 1_U)} U \xrightarrow{(1_U, U, f)} V) \\ f_*^* &= (V \xleftarrow{f} U \xrightarrow{1_U} U \xrightarrow{1_U} U) = (V \xleftarrow{(1_U, U, f)} U \xrightarrow{(1_U, U, 1_U)} U) \\ f^{**} &= (U \xleftarrow{1_U} U \xrightarrow{f} V \xrightarrow{1_V} V) = (U \xleftarrow{(f, U, 1_U)} V \xrightarrow{(1_V, V, 1_V)} V), \end{aligned}$$

where the middle diagrams are in $\mathcal{E}$ and the right diagrams are in $\mathbf{Spn}\mathcal{E}$.

We have the (bicategorical) adjunctions $f_{**} \dashv f_*^* \dashv f^{**}$ in $\mathbf{Ply}_{\text{lax}} \mathbf{Spn}\mathcal{E}$.

For each polynomial $p = (X \xleftarrow{r} A \xrightarrow{n} B \xrightarrow{t} Y)$ in $\mathcal{E}$, we have the diagram



in $\text{Spn}\mathcal{E}$ in which the diagonals are all bipullbacks. This proves that every polynomial p decomposes as

$$p \cong t_{**} \circ n^{**} \circ r_*^* \quad (3)$$

where $r_{**} \dashv r_*^*$, $n_*^* \dashv n^{**}$ and $t_{**} \dashv t_*^*$.

Tambara functors

Write $\text{Plyp}\mathcal{E}$ for the bicategory obtained from $\text{Ply}_{\text{lax}}\text{Spn}\mathcal{E}$ by taking isomorphism classes of 2-cells as 2-cells.

A *Tambara functor* on $\mathcal{E}$ is a CMon -enriched functor

$$T : \text{clPlyp}\mathcal{E} \rightarrow \text{CMon} .$$

An *objective Tambara functor* on $\mathcal{E}$ is a Cat_+ -enriched pseudofunctor

$$\mathbb{T} : \text{Plyp}\mathcal{E} \rightarrow \text{Cat}_+ .$$

Tambara as Mackey with structure

There is a $\mathbf{Cat}_+$ -enriched pseudofunctor

$$\Lambda : \mathbf{Spn}\mathcal{E} \longrightarrow \mathbf{Plyp}\mathcal{E}$$

taking the span $X \xleftarrow{u} S \xrightarrow{v} Y$ to the polynomial $X \xleftarrow{u} S \xrightarrow{1_S} S \xrightarrow{v} Y$. So every objective Tambara functor $\mathbb{T}$ restricts along Λ to give an objective Mackey functor $\mathbb{T}\Lambda$. Since Λ is the identity on objects, restriction along it is a monadic and comonadic pseudofunctor.

$$\mathcal{Tmb}\mathcal{E} = \mathbf{Ps}[\mathbf{Plyp}\mathcal{E}, \mathbf{Cat}_+]_+ \xrightarrow{\mathbf{Ps}[\Lambda, \mathbf{Cat}_+]_+} \mathbf{Ps}[\mathbf{Spn}\mathcal{E}, \mathbf{Cat}_+]_+ = \mathcal{Mky}\mathcal{E}$$

More abstractly we have a monadic and comonadic functor

$$\mathbf{Tmb}\mathcal{E} \longrightarrow \mathbf{Mky}\mathcal{E} .$$

My Review of the David Chan paper

The terminology here follows A. J. Blumberg and M. A. Hill [“Bi-incomplete Tambara functors”, London Math. Soc. Lecture Note Series **474** (Cambridge Univ. Press, 2022) 276–313; MR4327103] which built on their [“Incomplete Tambara functors”, Algebr. Geom. Topol. 18(2) (2018) 723–766; MR3773736].

For a finite group G , a Mackey functor M (in the form realised by Harald Lindner in 1976) is a finite-product-preserving functor with domain the category $\mathcal{A}^G = \text{Spn}(\text{set}^G)$ whose morphisms are isomorphism classes of spans $X \xleftarrow{r} A \xrightarrow{t} Y$ in the Boolean topos set^G of finite G -sets; the symbols r and t are to suggest “restriction” and “transfer”. For a couple of reasons, the codomain of M is taken, in this paper, to be the category Set of sets however each value MX is of course a commutative monoid since each object of $\mathcal{A}^G$ is; the condition that this monoid be an abelian group is imposed. The product in $\mathcal{A}^G$ is also the coproduct (and so it is the direct sum) and comes from the coproduct (disjoint union) in set^G .

The tensor product of Mackey functors is obtained by Day convolution using the monoidal structure on $\mathcal{A}^G$ coming from finite product in set^G . An incomplete Mackey functor has a domain with the spans restricted to those with t in some restricted class $\mathcal{O}$ of morphisms in set^G . The conditions on $\mathcal{O}$ (making it a so-called “indexing category”) imply that it contains all monomorphisms, is pullback stable, and is closed under composition, finite coproducts and finite products. The monoidal subcategory $\mathcal{A}_{\mathcal{O}}^G$ of $\mathcal{A}^G$ obtained by restricting to the spans with $t \in \mathcal{O}$ is then compact closed and so is an example of the situation studied by Elango Panchadcharam and the reviewer in [“Mackey functors on compact closed categories”, *Journal of Homotopy and Related Structures* **2(2)** (2007) 261–293].

A Tambara functor S for G is a finite-product-preserving functor with domain the category $\mathcal{P}^G$ whose morphisms are isomorphism classes of polynomials $X \xleftarrow{r} A \xrightarrow{n} B \xrightarrow{t} Y$ in set^G ; the symbol n is to suggest “norm”.

For the incomplete version, two indexing categories $\mathcal{O}_m$ and $\mathcal{O}_a$ are required to satisfy a compatibility in order for composition in $\mathcal{P}^G$ to restrict to $\mathcal{P}_{(\mathcal{O}_m, \mathcal{O}_a)}^G$ where the polynomials have $n \in \mathcal{O}_m$ and $t \in \mathcal{O}_a$. So $(\mathcal{O}_m, \mathcal{O}_a)$ -Tamara functors are defined with domain $\mathcal{P}_{(\mathcal{O}_m, \mathcal{O}_a)}^G$. The author provides an easily checked, combinatorial characterization of this compatibility.

Although I have not checked, it seems that the Hill and Hopkins [arXiv:1610.03114] concept of a symmetric monoidal $\mathcal{O}$ -Mackey functor $\mathcal{C}$ (see Definition 5.1 of the paper under review) is (in the terminology of Day and the reviewer [“Monoidal bicategories and Hopf algebroids”, Advances in Math. 129 (1997) 99–157; MR99f:18013]) a symmetric pseudomonoid in the symmetric monoidal bicategory $\text{Ps}_\times[\mathcal{A}_\mathcal{O}^G, \text{Cat}]$ of $\mathcal{O}$ -incomplete Mackey pseudofunctors (that is, objects are finite-product-preserving pseudofunctors from $\mathcal{A}_\mathcal{O}^G$ to the 2-category of categories and their tensor product is the appropriate convolution monoidal structure). A commutative monoid in $\mathcal{C}$ (Definition 5.6) is presumably a symmetric (lax-)monoidal morphism from the terminal monoidale to $\mathcal{C}$.

The author proves the Blumberg and Hill (2022) conjecture that a $(\mathcal{O}_m, \mathcal{O}_a)$ -Tambara functor is exactly a $\mathcal{O}_m$ -commutative monoid in the $\mathcal{O}_m$ -symmetric monoidal Mackey functor of $\mathcal{O}_a$ -Mackey functors.

Thank You



$$M \xrightarrow[\Delta]{\text{CoACT}} MQ$$